

A SRE-Dynamics Inspired Topological Paradigm for Composite Elementary Particles and Relational Space Emergence

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Resource & Availability Statement: This framework is built upon Status-Relational Entropy (SRE) Dynamics. The complete suite of theoretical materials is archived in the Zenodo open-data repository. **The full package includes system manuscripts, application developments, scientific hypotheses, complete algebraic derivations for operators 1-6, and simulation source code, all open-source.** Operators 7, 8, 9, 10 belong to subsequent closed-source commercial core modules and are not included in this document suite.

A Tencent Smart-Document workspace supporting AI-assisted review is available for both PC and mobile access.

As of 2026-08-14, the author no longer maintains or updates the Google Gemini Notebook SRE documentation suite due to Google Terms-of-Service constraints; this link serves purely as historical archive and shall not be used for formal citation:

- Gemini Notebook (historical archive, no longer updated):
<https://notebooklm.google.com/notebook/ef52bf5a-f6d0-4a2a-aed4-b25d6520ab2c>
- Tencent Smart-Document workspace:
<https://docs.qq.com/space/DUKRjYUtNWFdyV253>

According to the SRE principle, physical foundations originate from information statistics.

Reference baseline: SRE-v1.6 Axiom Suite (<https://doi.org/10.5281/zenodo.22077475>)

Historical references: earlier SRE archive DOIs retained for traceability.

Remark: This is a qualitative-semi-analytical topological hypothesis. Large-scale multi-degree-of-freedom numerical simulation and quantitative benchmarking against lattice-QCD are left for future follow-up research.

Ontological narrative note: The underlying physical driving force comes from the dissipation-compensation duality dynamics of SRE. The cross-spectral Hermitian matrix, coherence coefficient, condition number and related quantities are **statistical characterisation tools describing underlying dynamical behaviour, not ontological primitive operators generating physical effects.**

The original v1.0 manuscript contained causal inversion in narrative; this revised v1.1-rev version corrects the ontological causal hierarchy while keeping all mathematical formulas unchanged.

Abstract

Within the conceptual framework of Status-Relational-Entropy (SRE) Dynamics, this paper presents a qualitative-probabilistic formulation for the topological configuration and emergent geometry of composite elementary particles. Traditional physical paradigms rely heavily on fine-tuned continuous variables and empirical constants to explain rest-mass amplification and strong-interaction effects in composite structures. This work strips away all a-priori assumptions of absolute time, space and energy; space is de-indexed and

reformulated as a macroscopic geometric manifestation of status-relational entropy and phase coherence among distinct causal chains.

The underlying physical process is driven by **dissipation-compensation duality dynamics** of bidirectional Möbius causal loops. Coupled evolution of two internal causal-loops modifies mutual-information and phase-coherence between them. The 2×2 local cross-spectral Hermitian operator is a characterisation tool obtained by statistical smoothing over dynamical output time-series, capturing spectral-evolution features of the system. When the cross-coherence coefficient (at the representational level) asymptotically approaches unity, it corresponds to underlying dynamical saturation of phase coherence. The dimensionless eigenvalue-spacing of the ensemble spontaneously undergoes a distribution transition from the Wigner Surmise toward a continuous Poisson process. Within the full-rank expanded spectral regime, three-in-one co-emergent phenomena appear at the macroscopic rendering-layer: relational distance collapses toward zero due to maximised mutual information; the matrix condition-number hits numerical truncation thresholds, characterising an abrupt logical-pressure gradient rendered macroscopically as the strong-interaction force; instantaneous residual resonance triggers non-linear combinatorial explosion of secondary causal-feedback paths.

Free-parameter tuning and pre-existing background geometry are not required. This paradigm achieves self-consistent mathematical unification of composite-particle physics and relational metric space. Spectral-matrix constructions serve purely as statistical-analytical representations; the first-principle physical origin resides in dissipation-compensation duality evolution of the causal-information network.

[†]Note: In early manuscripts this fundamental resolution parameter was denoted Minimum Observational Step $\ell_{\min}$. Within the SRE-v1.6 axiom suite it is standardised as the **global evolution step ΔS** , representing the fundamental discrete state-refresh cycle of the causal network.

Keywords: Status-Relational Entropy; internal causal loops; composite elementary particles; relational space; dissipation-compensation duality; cross-spectral Hermitian operator; coherence saturation; gluon-sea-like mass amplification; strong interaction; Wigner-Poisson spectral-ensemble transition

1 Foundational Axioms: Single-Parameter System and Non-Background-Dependent Relational Space

1.1 Single-Parameter System and Local Counting of Time

The SRE architecture discards all macroscopic physical postulates. The rendered universe is derived from one single fundamental parameter — the **global evolution step $\Delta S^{\dagger}$** — which defines the absolute resolution threshold for causal operations. Within this logical framework:

- Time is strictly defined as local sequential counting of internal state-transitions within isolated causal chains, establishing a non-global intrinsic time axis.
- Rest-mass is neither an intrinsic material substance nor external computational overhead. It corresponds to topological depth and cumulative path-counts of internal causal-loops inside a node. When the global sorting protocol samples or attempts to displace a local node, it must fully traverse and process all internal feedback-paths contained within that node. This intrinsic structural complexity of the discrete graph manifests as inertial mass at the emergent layer.

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1.2 Relational-Distance Space as a Probabilistic Emergent Outcome

This framework strictly rejects any absolute background grid or pre-defined spatial coordinate indices (**non-background-dependent; no pre-existing spacetime manifold**). Space is a purely derived construct: macroscopic geometric rendering of status-relational entropy among decoupled causal chains.

The observed “relational distance” between two causal nodes at the macroscopic layer represents discrete information impedance and sequential step-delay required for step-size cross-correction between nodes at the underlying protocol layer.

1.3 Local Cross-Spectral Operator and Information-Geometric Mapping

Ontological remark: $X_0(t)$ and $X_1(t)$ are discrete complex time-series samples output by underlying dissipation-compensation duality dynamics. The 2×2 local complex cross-spectral Hermitian matrix $\mathbf{M}$ below is a **statistical characterisation matrix** obtained by local statistical smoothing over dynamical outputs. It describes system-evolution features and is not an ontological primitive operator generating physical phenomena.

Let $X_0(t)$ and $X_1(t)$ denote discrete complex response streams output by dynamical evolution of two underlying internal causal-loops. Within the generalised spectral domain, local statistical smoothing over an iterative window yields the 2×2 local complex cross-spectral Hermitian operator $\mathbf{M}$:

$$\mathbf{M} = \begin{pmatrix} E[|X_0|^2] & E[X_0 X_1^*] \\ E[X_1 X_0^*] & E[|X_1|^2] \end{pmatrix}$$

The modulus of the non-linear cross-coherence term $\rho = |E[X_0 X_1^*]| \in [0, 1]$ serves as a characterisation quantity directly quantifying logical correlation density between two causal chains. Emergent macroscopic distance D is not propagated from an external background; it is derived strictly from Shannon mutual-information encoded within the local Hermitian-matrix representation. Define shared status-relational entropy between two loops as $I = -\ln(1 - \rho^2)$. Emergent geometric distance is inversely proportional to this shared logical density:

$$D \propto \frac{1}{I} = \frac{1}{-\ln(1 - \rho^2)}$$

- When $\rho \rightarrow 0$, mutual-information vanishes ($I \rightarrow 0$), emergent distance $D \rightarrow \infty$. The two causal chains behave as fully decoupled, independent, infinitely separated particles.
- When underlying dissipation-compensation duality dynamics drive causal-loops toward phase-coherence saturation — represented at the characterisation level by $\rho \rightarrow 1$ — mutual-information saturates toward $I \rightarrow \infty$ and forces emergent distance $D \rightarrow 0$. Two causal-loops spatially overlap completely. Physical “approach” or “contact” of particles is an emergent manifestation of relational-metric-space collapse following underlying causal-network phase-coherence saturation; ρ is merely a spectral indicator for this physical process.

Using single-pass first-order closed-form algebra, raw eigenvalue spacing $\Delta\lambda$ and local condition number κ are extracted; both quantities are spectral characterisers of underlying dynamical behaviour:

$$\Delta\lambda = \sqrt{\text{Tr}(\mathbf{M})^2 - 4 \det(\mathbf{M})}$$

$$\kappa = \frac{\text{Tr}(M) + \Delta\lambda}{\max(\text{Tr}(M) - \Delta\lambda, \varepsilon)}$$

$$\varepsilon = 10^{-7*}$$

*Note: $\varepsilon = 10^{-7}$ is purely a numerical regularisation cutoff for matrix-computation pipelines. **It is not a fundamental physical constant of the underlying causal-information network.**

2 Core Paradigm: Three-in-One Emergent Jump and Composite-Particle Birth

Ontological remark: Numerical change in ρ does **not** drive underlying physics. Instead: **underlying bidirectional Möbius causal-loops are driven toward phase-coherence saturation by dissipation-compensation duality dynamics**. This physical dynamical process manifests spectrally as $\rho \rightarrow 1$, which in turn causes full-rank expansion $\det(M) > 0$ of the local cross-spectral matrix M , triggering radical redistribution inside the probabilistic feature-space.

2.1 Spectral Transition from Wigner to Poisson Ensemble

Once a full-rank coupled physical state is established by underlying dynamics, at the representational generalised spectral-ensemble level, the dimensionless ensemble-spacing metric $s = \frac{\Delta\lambda}{E[\Delta\lambda]}$ spontaneously transitions from isolated Wigner-Surmise statistics to a continuous Poisson-process: $P(s) = e^{-s}$ (valid for global sequentially-sorted generalised spectral-ensembles).

Under Poisson-clustering ($P(s) \rightarrow 1$ as $s \rightarrow 0$), microscopic eigenvalue-spacing compresses densely toward the centre. Algebraically the denominator term $(\text{Tr}(M) - \Delta\lambda)$ is forced toward the numerical cutoff limit $\varepsilon = 10^{-7}$. This phenomenon is the spectral representational signature of underlying causal-network dynamics.

2.2 Three-in-One Co-Emergence of Space, Force and Mass

Defining space as a relational emergent construct implies composite-particle formation is an algebraic necessity of underlying causal-network evolution. Under dissipation-compensation-driven phase-coherence saturation (represented by $\rho \rightarrow 1$), three inter-connected phenomena spontaneously appear at the macroscopic layer:

- 1. Relational spatial overlap ($D \rightarrow 0$):** Underlying dissipation-compensation dynamics produce cross-coherence saturation and maximised mutual-information. Macro-geometric distance collapses toward zero at the representational level; two independent loops merge into a single local composite node possessing internal topology.
- 2. Emergence of interaction force ($\kappa \rightarrow \infty$):** Underlying causal information-streams mutually modify and “chase” each other’s discrete step-sizes, generating intense logical-pressure gradients. This dynamical effect manifests spectrally as eigenvalue-spectrum compression and multi-order logarithmic spike of condition-number κ , rendered macroscopically as the strong-interaction force. No mechanical force constants are manually introduced.
- 3. Rest-mass amplification: combinatorial explosion of secondary causal-feedback paths:** In isolation, a single causal-loop processes only its intrinsic $2N$ steps. Within the underlying full-rank-coupled physical phase-space, maintaining zero-residual global-protocol consensus across the unified node causes steps from loop X_0 to continuously trigger and cross-correct states of X_1 . This

triggers combinatorial explosion of intertwined secondary causal-feedback paths at the discrete-graph layer.

Total cumulative loop-steps and path-depth of the composite entity transitions from linear summation toward high-order graph-ensemble mapping, exhibiting qualitative scaling bifurcation:

$$\text{Total Causal Paths} = \int_0^\infty \kappa(s) \cdot f(\text{Secondary Feedback Path Generations}) \cdot e^{-s} ds$$

Note: This equation evaluates topological scaling-bifurcation using spectral-representational quantities; it is not a numerically-fitted expression. Driven by condition-number-spike high-frequency step-correction near the cutoff boundary, the discrete causal-graph undergoes structural phase-transition with exponential complexity growth. It provides a self-consistent, non-fitted mathematical account for step-wise non-linear rest-mass amplification observed upon sub-loop binding.

3 Geometric Pruning of Internal Loops via Instantaneous Path Alignment

Static phase-rotations of the complex Hermitian matrix M leave time-averaged eigenvalues invariant; this is merely a mathematical property at the representational level. True system dynamics originate from evolution of underlying bidirectional causal-loops. Instantaneous state-trajectories within discrete statistical-smoothing windows are strictly governed by phase-locking-alignment mechanisms intrinsic to the dual-loop ontology. The macroscopic concept of parallel / anti-parallel spin can thereby be fully decomposed into purely discrete graph-path mechanisms:

- 1. Anti-phase alignment (opposite-spin mode):** Under discrete beats, Möbius residuals of the dual loops point in opposing directions within the complex plane. Underlying dynamics produce high-frequency algebraic cancellation of instantaneous residuals prior to window-smoothing, suppressing physical divergence and preventing representational-matrix condition-number from reaching critical divergence. Secondary-feedback-path generation remains stable and bounded, explaining why certain two-loop configurations yield low-order mass profiles.
- 2. In-phase alignment (parallel-spin mode):** Möbius residuals align symmetrically, triggering constructive phase-resonance at the physical level. Severe full-rank deformation of underlying loops occurs, forcing protocol-layer high-intensity local-pruning functions $(1.0 - \alpha e^{-s})$ at every beat to maintain topological closure. This triggers structural combinatorial explosion of secondary causal paths, manifesting as higher-mass structural configurations.

Discussion

The composite-particle-generation picture advanced in this paper constitutes a **qualitative-semi-analytical topological hypothesis via structural homomorphic analogy**. Homomorphic analogy means the model preserves core structural relations of underlying causal-network coupling-evolution, yet partial microscopic details may be lost when mapping onto real QCD hadron systems.

Model Scope and Boundaries

1. Achievable goals (mechanism level)

Built upon SRE underlying dissipation-compensation-duality dynamics, this model delivers a conceptually self-consistent mechanistic explanation: when two internal causal-loops approach phase-coherence saturation, combinatorial explosion of secondary causal-feedback-paths produces

non-linear amplification of rest-mass. This topological picture addresses the physical mechanism: *why mass increases significantly upon binding*. It also delivers a unified ontological account for the co-emergence of relational-space collapse, strong-interaction emergence, and spin phase-alignment phenomena.

2. Current limitations (quantitative level)

⚠ At present this model **cannot derive real-world gluon-sea mass-amplification factors, absolute hadron-mass values, nor precise baryon-spectral mass-splitting quantitative results**.

- The total-causal-paths integral shown above

$$\text{Total Causal Paths} = \int_0^\infty \kappa(s) \cdot f(\text{Secondary Feedback Path Generations}) \cdot e^{-s} ds$$

is a **qualitative formal expression characterising topological scaling bifurcation**, not a complete quantitative formula ready for direct GeV-scale numerical evaluation.

- The 2x2 simplified cross-spectral Hermitian operator represents a two-loop toy-model only. Real hadron-systems are complex networks with huge numbers of coupled degrees-of-freedom; this model performs degree-of-freedom reduction. The homomorphic mapping preserves structural evolutionary logic, yet a complete quantitative conversion bridge between SRE topological quantities and QCD / lattice-QCD observables has not yet been established.
- Neither $\kappa(s)$ nor the secondary-feedback-path-generation function $f(\cdot)$ have been fully analytically determined from SRE-v1.6 axioms. Calibration constants required for matching against experimental / lattice-QCD datasets are absent. Therefore mass-amplification factors for real-world protons, neutrons and other hadrons cannot be output presently.

Further constrained by SRE axiom architecture: the framework prescribes only the global evolution-step ΔS as minimal causal-operation resolution limit, **but does not pre-assign a fixed, precise underlying microscopic topological-connectivity structure. Local sub-graph connectivity configurations themselves are emergent outcomes of dissipation-compensation-duality dynamics and are not a-priori axiomatic inputs**.

From this we obtain a robust qualitative mechanistic inference: when two sets of internal causal-loops reach phase-coherence saturation, combinatorial explosion of secondary causal-feedback-paths inevitably produces non-linear mass-surge for composite systems.

Nevertheless, the final total number of secondary feedback-paths generated depends sensitively upon real-time local topological-details evolved by dynamics. Given there exists no pre-specified fixed microscopic topology, the minimal-resolution scale ΔS alone is insufficient to uniquely constrain path-proliferation magnitude. Consequently purely from axioms alone, deterministic analytical derivation of unique mass-amplification-factors or absolute hadron-masses is impossible.

This contrasts with lattice-QCD: its numerical calculations rest upon pre-defined fixed discrete lattices plus SU(3) gauge-degree-of-freedom bases. Within the SRE picture topological connectivity itself is dynamical output rather than computational input. Progress toward quantitative outputs requires large-scale network-simulations allowing spontaneous emergence of local topology, followed by statistical extraction of observables; such work lies outside scope of this qualitative-semi-analytical mechanistic paper.

Comparison with Quantum Chromodynamics (QCD / Lattice-QCD)

It is instructive to compare against established quantum-chromodynamics: the SU(3) gauge Lagrangian of QCD serves as its underlying axiom, yet no closed-form analytical solution exists for the low-energy confinement regime. Pure algebraic derivation yielding gluon-sea mass-amplification factors is unavailable numerically. Present-day numerical hadron-masses and gluon-contribution fractions are outputs from large-scale non-perturbative lattice-QCD simulations; these are **outputs of first-principle numerical simulation rather than closed-form analytical derivations**.

This SRE model operates at an ontological level more fundamental than QCD degrees-of-freedom. The 2-loop toy-model here delivers a qualitative-semi-analytical homomorphic picture revealing the mechanism: coupling-phase-coherence-saturation triggers secondary-feedback-path-explosion giving non-linear rest-mass amplification. The total-causal-paths integral is only a formal expression characterising topological scaling bifurcation. Secondary-feedback-generation $f(\cdot)$ and condition-number function $\kappa(s)$ are not fully solved analytically from SRE-v1.6 axioms. Therefore real-world gluon-sea mass-amplification factors cannot be output at present. To achieve quantitative results in future requires large-multi-degree-of-freedom network-simulation plus construction of complete quantitative-mapping bridges between SRE topological quantities and QCD observables.

Future direction: AI-assisted inverse inference from sampled datasets

Even though pure forward analytical derivation of deterministic mass-amplification factors cannot be achieved solely from SRE axioms, one post-processing computational pathway remains open for future investigation: inverse-problem identification via large-sample datasets combined with machine-learning / AI-driven regression.

From inverse-problem perspective: SRE-v1.6 axioms define dynamical evolution rules (dissipation-compensation duality, minimal resolution given by global evolution-step ΔS). However local topological-connectivity is dynamical output instead of pre-defined input, leaving degrees-of-freedom not uniquely locked by axioms alone.

If a large-multi-degree-of-freedom forward SRE causal-network simulator is constructed in future, large batches of coupled-loop system simulation-samples can be generated, pairing underlying topological-statistical-features against emergent macroscopic observables. Real-world observational samples from lattice-QCD or experimental hadron-spectra may also be incorporated. Based on this sample library, Bayesian inference, sparse-system-identification or physics-informed-machine-learning can perform inverse inference: constraining secondary-feedback-generation function $f(\cdot)$ and condition-number function $\kappa(s)$ from macroscopic observables, establishing effective mappings between topological statistics and hadronic observables.

Important distinction: this paradigm **is not equivalent to pure axiomatic first-principle analytical derivation**. Axioms supply dynamical rules, yet sample datasets play an essential role constraining degenerate solution-space. This inverse-problem intrinsically carries risks of solution-degeneracy: distinct underlying topological-configurations may yield similar macroscopic observables. Strong SRE ontological priors must be embedded to shrink degenerate solution-space. Prediction performance also strongly depends upon completeness of sample-configuration-coverage; extrapolation toward unseen novel configurations carries failure-risk.

In summary: AI-sample-driven inverse identification represents a promising future computational pathway; it is outside scope of this manuscript. This paper only elaborates the qualitative-semi-analytical mechanism that phase-coherence-saturation yields non-linear mass-surge.

Follow-up research roadmap

To move beyond homomorphic-analogy-only mechanism description three follow-up tasks are required:

1. Extend the 2-loop toy-model toward large-multi-degree-of-freedom causal-network simulation, escaping two-body simplifying assumptions.
2. Build complete mapping rules converting SRE topological quantities (path-counts, condition-numbers, feedback-path proliferation) into QCD observables (energy-momentum-tensor, hadron rest-masses).
3. Calibrate against public lattice-QCD datasets to determine concrete functional forms for $f(\cdot)$ and other unknown functions, enabling predictions for real-world mass-amplification-factors and hadron-spectra.

Until completion of above tasks, all results within this paper shall be strictly confined to

qualitative-semi-analytical ontological-mechanism discussion; no numerical predictions for hadron-masses are made.

Conclusions

As pointed-out within the Discussion section, the present paradigm is a qualitative-semi-analytical topological picture via structural homomorphic analogy. It can reveal internal mechanisms for composite-particle formation but cannot output quantitative physical values such as real-world gluon-sea mass-amplification-factors.

This formulation advances SRE Dynamics toward a fully non-background, loop-intrinsic paradigm. The structural validity of composite elementary particles is no longer an isolated question of material composition. Instead it emerges as a unified probabilistic outcome: driven by dissipation-compensation-duality of underlying bidirectional Möbius causal-networks reaching phase-coherence-saturation, giving co-occurrence of relational-space collapse, interaction-force manifestation and secondary-loop-path proliferation.

Cross-spectral matrices, condition-numbers and coherence-coefficients are merely statistical-characterisation and analytical tools for describing this evolutionary process. **Fundamental physical properties of composite particles are rigid inevitable outcomes of dissipation-compensation-duality dynamics of the underlying causal-information-network; they are not produced by matrix-operators themselves.** This mathematical model frees itself from empirical physical constraints and fixed spacetime backgrounds.

Supplementary remark: This manuscript delivers qualitative-semi-analytical topological insight for gluon-sea-like mass-amplification inside composite-particles. Full quantitative numerical validation against hadron-spectra is reserved for follow-up research.

References

1. SRE-Dynamics Axiom Suite v1.6, Zenodo archive.
2. SRE early-archive series DOIs for traceability.
3. Lattice QCD: research on gluon-sea mass-generation inside hadrons.